

Path Integral Monte Carlo for the Quantum Harmonic Oscillator and the λy^4 Interaction

Federico G. Malara (668332)
University of Pisa

A.Y. 2025/2026

Abstract

The quantum harmonic and anharmonic (λy^4) oscillators are studied by means of Path Integral Monte Carlo on a Euclidean lattice. Thermodynamic observables and the energy spectrum are reconstructed through two-point correlators, and the results extrapolated to the continuum are compared with analytical solutions and exact diagonalization within statistical uncertainties.

1 Monte Carlo Choices

The numerical simulations were carried out by combining a local heatbath update with $n_{\text{over}} = 5$ microcanonical overrelaxation steps applied to all sites of the lattice [1]. In the purely harmonic case, the localized update is governed by an exact Gaussian conditional distribution, which reduces the autocorrelation time scales as documented in App. A. Based on the asymptotic relaxation study of the slowest observable $\langle y \rangle$, the initial thermalization transient was removed by discarding the first $n_{\text{therm}} = 4 \times 10^5$ sweeps [2].

2 Thermodynamics of the Harmonic Oscillator

The quantum harmonic oscillator is formulated through the Euclidean path integral discretized on a periodic temporal lattice of extent β and dimensionless step $\eta = a\omega = \beta/N_t$ (in natural units $\hbar\omega = 1$). The resulting lattice action takes the form [1, 3]:

$$S_E^{(L)}[y] = \sum_{j=0}^{N_t-1} \left[y_j^2 \left(\frac{\eta}{2} + \frac{1}{\eta} \right) - \frac{1}{\eta} y_j y_{j+1} \right], \quad (1)$$

where y_j denotes the spatial coordinate rescaled with respect to the characteristic length $\ell = \sqrt{\hbar/(m\omega)}$. The Monte Carlo estimates were validated by comparison with the analytical finite-temperature internal energy:

$$\langle y^2 \rangle_{\text{exact}} = \langle H \rangle_{\text{exact}} = \frac{1}{2} \coth \left(\frac{\beta}{2} \right). \quad (2)$$

For the reference thermal extent $\beta = 5$, the exact asymptotic value is $0.50678365 \dots$ [1].

2.1 Euclidean Paths

The effect of lattice discretization on the resolution of quantum paths is illustrated in Fig. 1. Physically, the harmonic potential constrains large-scale macroscopic fluctuations, while the kinetic contribution determines the local microstructure (ultraviolet fluctuations) as N_t increases [1, 3].

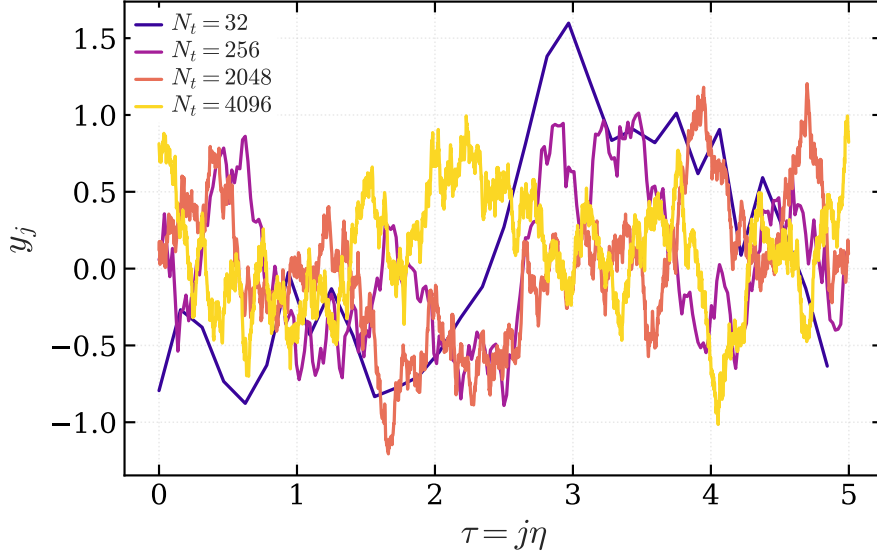


Figure 1: Euclidean path configurations y_j at fixed temporal extent $N_t\eta = 5$. Reducing the step η increases the resolution of local ultraviolet fluctuations.

2.2 Continuum Extrapolation

The renormalized internal energy of the system was determined by combining the spatial and kinetic estimators:

$$H_{\text{ren}} = \frac{1}{2}\langle y^2 \rangle + \frac{1}{2\eta} - \frac{1}{2\eta^2}\langle (\Delta y)^2 \rangle, \quad (3)$$

where $\langle (\Delta y)^2 \rangle = \frac{1}{N_t} \sum (y_{j+1} - y_j)^2$ quantifies the lattice gradient. The analytical counter-term $1/(2\eta)$ is essential to remove the ultraviolet divergence that arises in the continuum limit $\eta \rightarrow 0$ [1, 3].

The systematic corrections induced by lattice discretization follow an asymptotic behavior of the form $a + b\eta^2$. In order to isolate the dominant scaling regime and identify the upper cutoff η_{max} , the normalized deviation $z(N_t) = (\langle y^2 \rangle_{N_t} - \langle y^2 \rangle_{\text{exact}})/\sigma_{N_t}$ was monitored. As shown in Fig. 2, the operational threshold $|z| \gtrsim 3$ justifies the exclusion of points with $\eta > 0.2$, where systematic lattice effects dominate over statistical fluctuations.

The stability of the fits was tested through a systematic scan of the lower cutoff η_{min} by monitoring the evolution of χ_{red}^2 (Fig. 4), avoiding contamination from critical slowing down at excessively small steps. The final extrapolation fits for the mean squared position and for the internal energy are shown respectively in Fig. 3a and Fig. 3b. The continuum intercept yields $a = \langle y^2 \rangle_{\text{cont}} = 0.5060(8)$ with $\chi_{\text{red}}^2 = 0.887$ and $a = H_{\text{ren,cont}} = 0.5053(22)$ with $\chi_{\text{red}}^2 = 0.900$, both in statistical agreement with the theoretical limits.

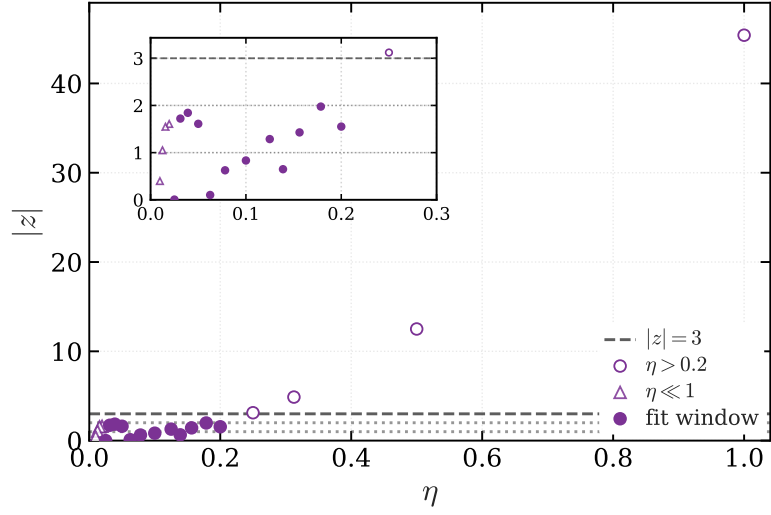


Figure 2: Normalized absolute deviation $|z|$ of $\langle y^2 \rangle$ as a function of η . The horizontal lines highlight deviations of 1σ , 2σ , and 3σ of the Monte Carlo error. The points with $|z| > 3$ justify the upper limit $\eta_{\max} = 0.2$, beyond which the systematic discretization error dominates over the statistical fluctuations [1].

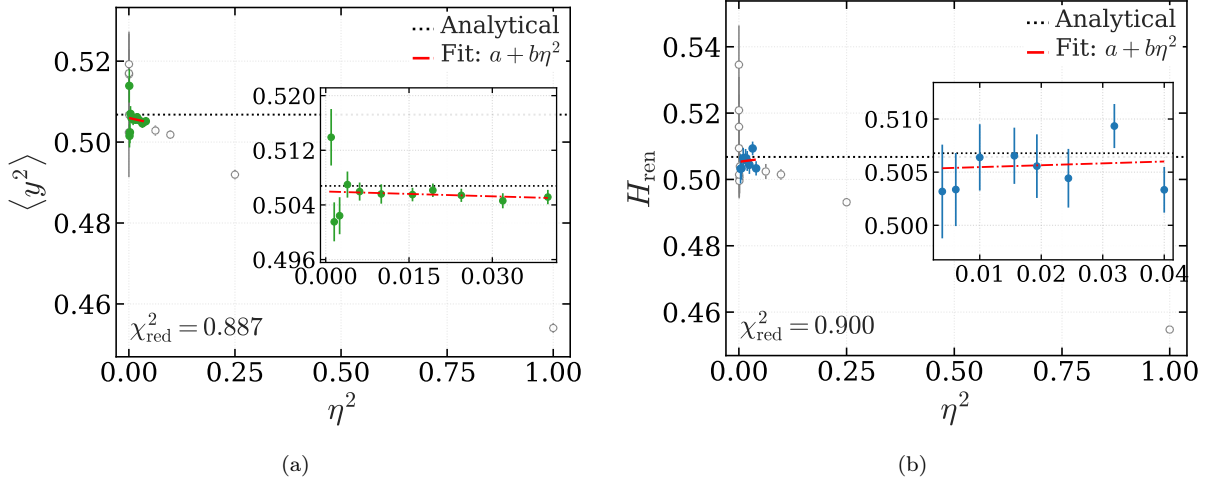


Figure 3: Continuum extrapolation: (a) $\langle y^2 \rangle$ at $\beta = 5$ as a function of η^2 ; (b) renormalized energy H_{ren} at $\beta = 5$ as a function of η^2 . The dash-dotted red line describes the fit over the stable points.

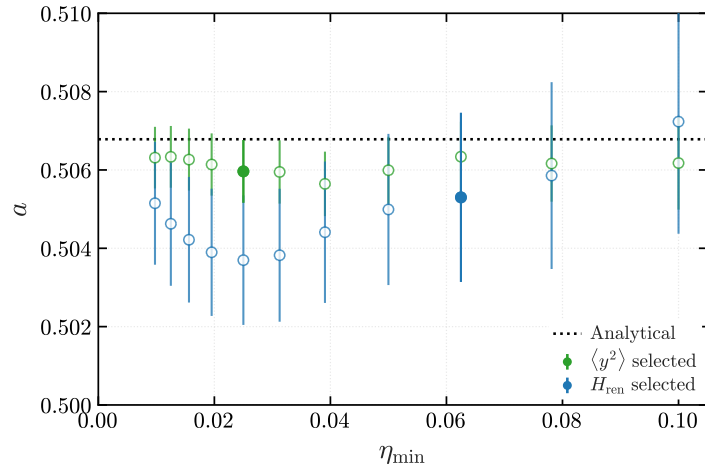


Figure 4: Stability study of the intercept a as $\eta_{\min}$ is varied at fixed $\eta_{\max} = 0.2$.

2.3 Virial Theorem

The quantum energy balance was verified by separating the potential energy $V = \frac{1}{2}\langle y^2 \rangle$ from the renormalized kinetic estimator $K_{\text{ren}} = K_{\text{div}} + 1/(2\eta)$, where $K_{\text{div}} = -\langle (\Delta y)^2 \rangle / (2\eta^2)$. The virial theorem imposes the asymptotic identity $K_{\text{ren}} = V$ in the continuum limit [1, 3].

As shown in Fig. 5a, the unrenormalized observable K_{div} exhibits the expected negative ultraviolet divergence, mirrored by the growth of the counterterm [3]. The direct comparison between the two energy components (Fig. 5b) confirms the full validity of the theorem within the sampled statistical uncertainties.

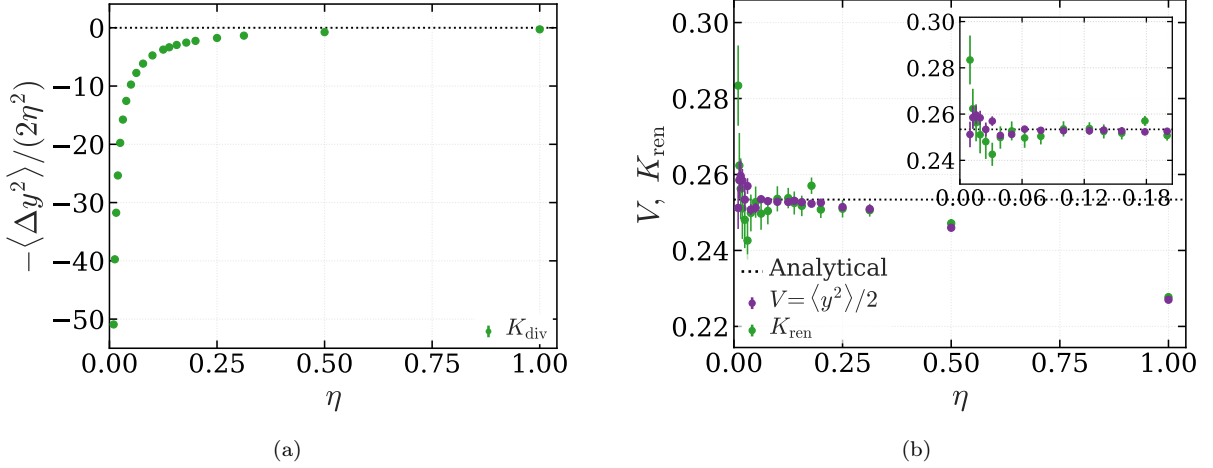


Figure 5: Analysis of the kinetic estimator at $\beta = 5$: (a) asymptotic ultraviolet divergence of K_{div} as a function of η . (b) Verification of the virial theorem, within errors, through direct comparison between V and K_{ren} .

2.4 Temperature Dependence

The thermal response of the system was isolated by working at fixed lattice step ($\eta = 0.05$) in order to freeze the value of the divergent counterterm. Defining the energy difference with respect to the lowest-temperature point ($\beta_0 = 20$, taken as an approximation of the ground state), the ultraviolet constant cancels exactly [3]:

$$\Delta U_b(\beta) = U_b(\beta) - U_b(\beta_0). \quad (4)$$

The results of this procedure are summarized in Fig. 6, where the macroscopic energy difference is compared with the expected distribution $n_B(\beta) = 1/(e^\beta - 1)$.

The weighted interpolation of the data (Tab. 1) using the functional form

$$\Delta U_b(x) = A [n_B(x^{-1}) - n_B(x_0^{-1})] \quad (5)$$

yields an amplitude $A = 0.989(12)$ with $\chi_{\text{red}}^2 = 0.735$, confirming a good extraction of the purely thermally excited component of the system.

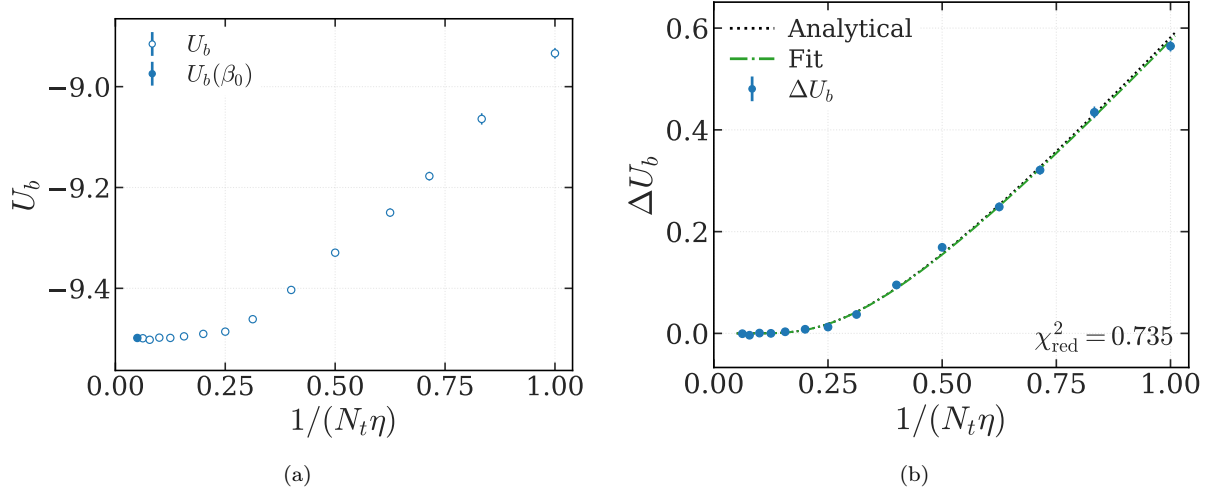


Figure 6: Thermal dependence at $\eta = 0.05$: (a) internal energy U_b as a function of β^{-1} . (b) Fit of the energy difference ΔU_b .

η	β_0	A	χ_{red}^2
0.05	20	0.989(12)	0.735

Tabella 1: Parameters extracted from the thermal fit of ΔU_b .

3 Euclidean Correlators and Energy Gaps

The energy spectrum of the system was determined by analyzing the asymptotic behavior of two-point correlators in Euclidean time. For a generic observable $\hat{O}$ with a nonzero vacuum expectation value, the connected correlator is defined as:

$$C_O^{(c)}(\tau) = \langle O(\tau)O(0) \rangle - \langle O \rangle^2. \quad (6)$$

In the low-temperature limit ($\beta \rightarrow \infty$), the function admits the spectral decomposition:

$$C_O^{(c)}(\tau) \simeq \sum_{k>0} |\langle 0|\hat{O}|k \rangle|^2 e^{-(E_k - E_0)\tau}, \quad (7)$$

where the large-time decay is dominated by the smallest energy gap among those geometrically coupled by the operator [3].

In order to extract the energy spectrum from the sampled low-temperature data ($\beta = 40$), two distinct methodological approaches were implemented and compared: a local analysis based on the ratio between contiguous time points and a global analysis based on nonlinear fits extended to the full temporal profile of the curves.

Local approach: Correlator ratio. On the discretized lattice with step η , the local gap at time step t is extracted by computing the logarithm of the ratio between the fluctuations at contiguous times:

$$\Delta(t) = \frac{1}{\eta} \log \frac{C^{(c)}(t\eta)}{C^{(c)}((t+1)\eta)}. \quad (8)$$

The samples were obtained on four lattices ($N_t = 60, 80, 100, 200$), with uncertainties propagated via jackknife. The linear operator y and the connected correlator of y^2 made

it possible to isolate the first two gaps, $\Delta_1 = E_1 - E_0$ and $\Delta_2 = E_2 - E_0$ [1, 3]. By contrast, the y^3 channel is contaminated at long times by the lower-energy odd state (Δ_1). To remove this spurious overlap and isolate the third gap (Δ_3), the optimized operator $A = y^3 - \frac{3}{2}y$ was implemented [1, 3].

The time dependence of the local gaps is illustrated in Fig. 7. To assess the impact of discretization effects without introducing free fit parameters, Fig. 8 reports the direct deviations of the gaps extracted from the plateau with respect to the exact analytical values, $\Delta(\eta) - \Delta_{\text{exact}}$, as functions of η^2 . The regular alignment of the points shows an ordered convergence toward the zero line as the lattice step decreases, visually confirming the compatibility of the stochastic data with the expected quadratic scaling. The numerical results associated with this first local methodology are reported in Tab. 2.

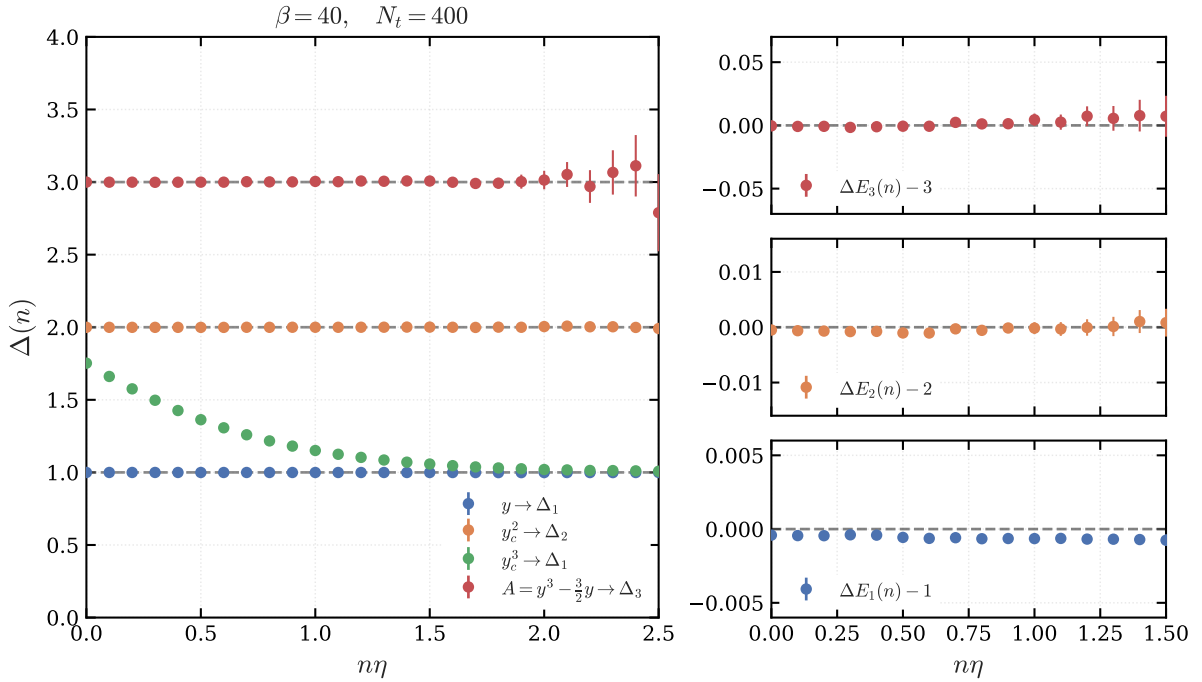


Figure 7: Estimation of the harmonic oscillator energy gaps through the local-gap analysis for $\beta = 40$ and $N_t = 400$. (a) Shows the quantity Δ for the operators y , y^2 , y^3 , and A . (b) Shows the deviations from the exact values.

Correlator	Exact value	Δ_k
C_y	1	0.99973(26)
$C_{y_{\text{conn}}^2}$	2	1.99996(62)
C_A	3	3.00009(15)

Tabella 2: Harmonic-oscillator energy gaps extracted through the local plateau analysis at $\beta = 40$. Jackknife errors.

Global approach: Hyperbolic-cosine fit. To exploit the full statistical profile of the curves and minimize fluctuations related to the choice of the plateau window, a global fit procedure was implemented on the time dependence of the connected Euclidean correlators. The theoretical functional form assumes an exact hyperbolic-cosine

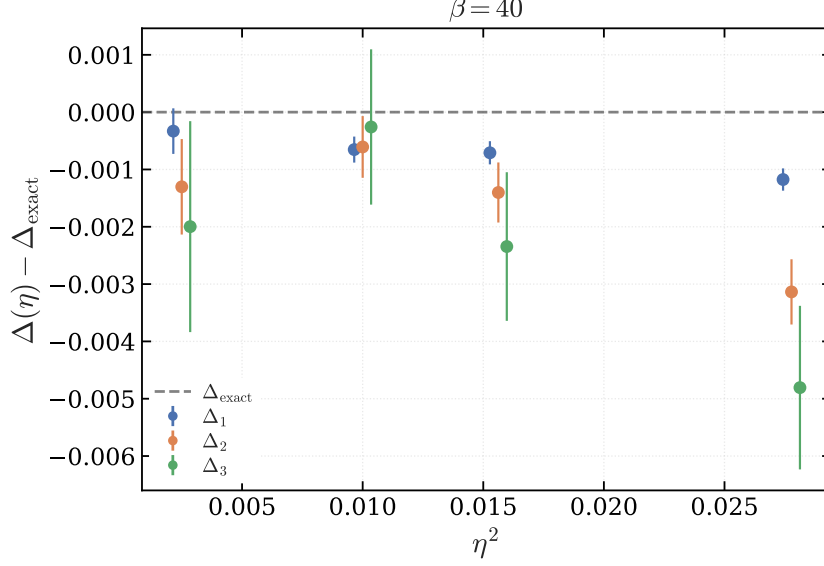


Figure 8: Scaling of the deviations of the first three harmonic-oscillator energy gaps as a function of η^2 . The horizontal line identifies the theoretical reference zero.

symmetry:

$$C_O^{(c)}(\tau) = A_O [e^{-\Delta_k \tau} + e^{-\Delta_k(\beta-\tau)}]. \quad (9)$$

The nonlinear interpolation through Eq. (9) was performed in $\eta \in [0.05, 0.80]$. The behavior of the normalized connected correlator of the fundamental channel is illustrated in Fig. 9a, showing the regular convergence toward the hyperbolic-cosine fit curve as the lattice step is refined. The extrapolation in η^2 for Δ_1 is reported in Fig. 9b.

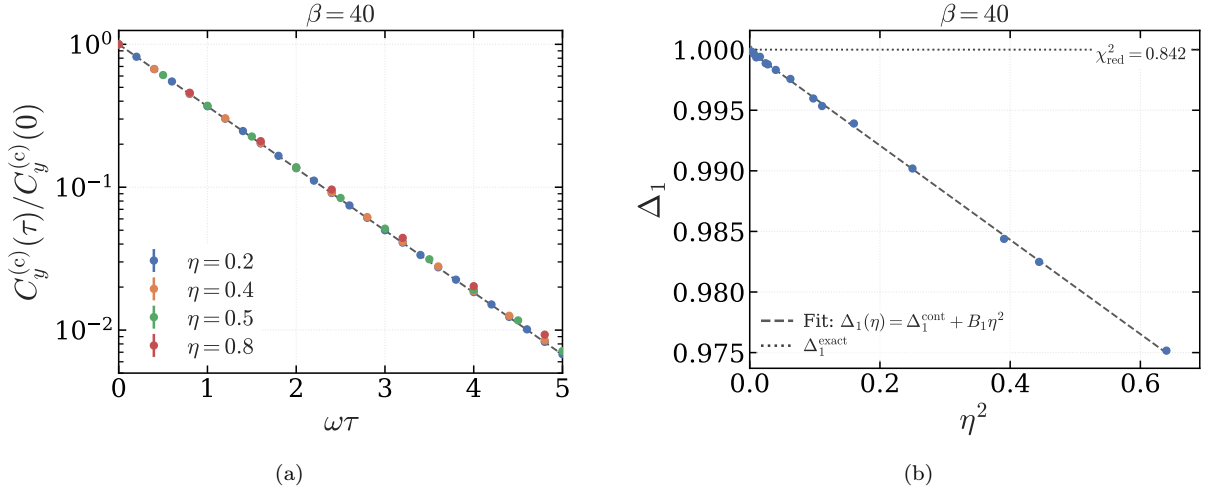


Figure 9: Global analysis of the fundamental gap at $\beta = 40$: (a) decay of the normalized connected correlator as η varies; (b) continuum extrapolation of the gap Δ_1 .

For the third gap Δ_3 , the coarser lattices yielded larger noise in the fit. To obtain physical consistency, a geometric cutoff on η was imposed by restricting the fit window.

The resulting continuum extrapolations for the excited channels Δ_2 and Δ_3 from the global hyperbolic-cosine fits are shown respectively in Fig. 10a and Fig. 10b.

The final parameters extracted through this second global methodology are collected analytically in Tab. 3.

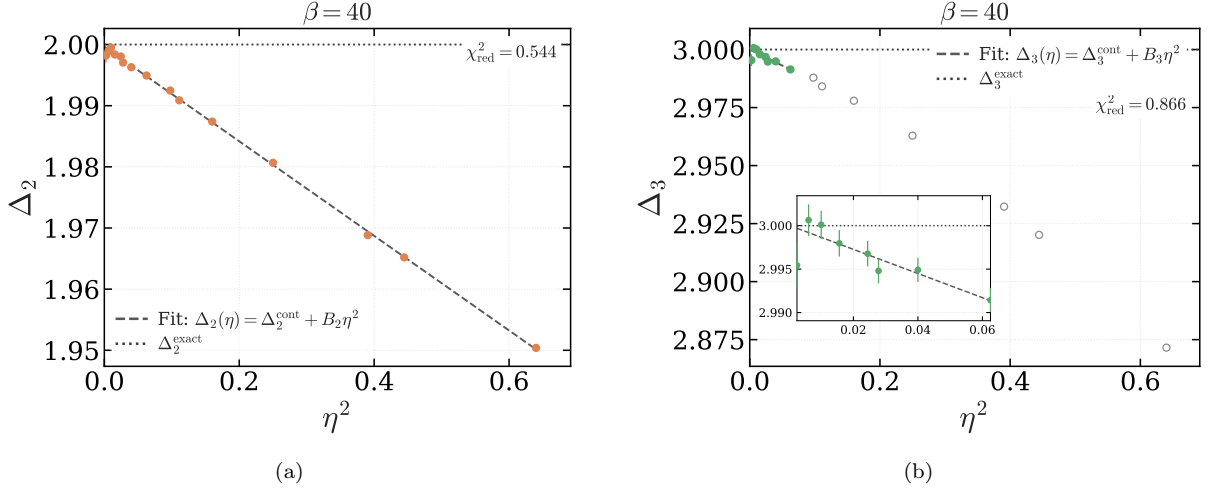


Figure 10: Continuum extrapolations of the excited gaps through global cosh fits as functions of η^2 : (a) second gap Δ_2 ; (b) third gap Δ_3 computed on the subset $\eta \leq 0.25$.

Operator	Gap	Δ_k^{cont}	χ_{red}^2
y	Δ_1	0.999871(67)	0.842
y_{conn}^2	Δ_2	1.999644(185)	0.544
A	Δ_3	3.00005(100)	0.866

Tabella 3: Energy spectrum of the harmonic oscillator obtained from the global hyperbolic-cosine fit analysis and extrapolated to the continuum at $\beta = 40$. Jackknife errors.

3.1 Spatial Distribution and Classical Crossover

The spatial probability density of the particle, $P_\beta(y) = \langle \delta(q - y) \rangle_\beta$, was determined by accumulating the visit frequency of the coordinate along the Euclidean paths [3]. As illustrated in Fig. 11(a), at low temperature ($\beta = 40$) thermal excitations are suppressed and the distribution collapses onto the squared modulus of the ground-state wave function, $|\psi_0(y)|^2$ [3]; consequently, $\langle y^2 \rangle_{T=0} = 1/2$.

As the temperature increases, the spatial distribution broadens progressively until it converges to the classical Boltzmann statistics.

Fig. 11(b) shows the behavior of the variance: saturated at $1/2$ by zero-point motion at low temperatures, it converges to the $1/\beta$ behavior for $\beta \ll 1$, where the product $\beta \langle y^2 \rangle$ tends to unity.

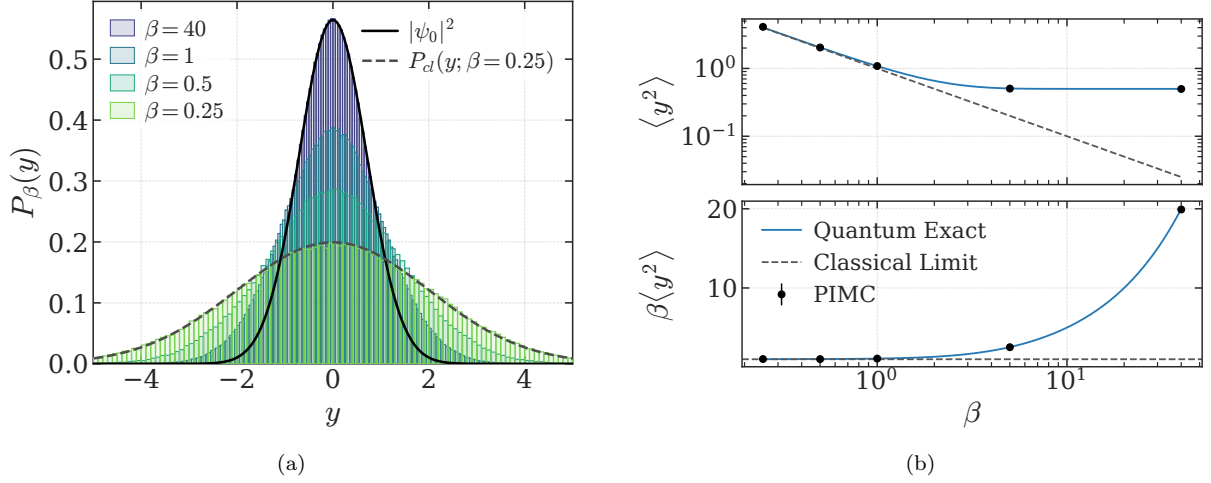


Figure 11: (a) Probability density confined between the quantum ground state and the classical Boltzmann curve. (b) Variance $\langle y^2 \rangle$ and rescaled quantity $\beta \langle y^2 \rangle$, highlighting the quantum saturation at low temperature and the recovery of the classical asymptotic limit.

4 Extension with λy^4 Coupling

The departure from the harmonic regime was studied by introducing a quartic perturbation in the dimensionless potential of the system:

$$V(y) = \frac{1}{2}y^2 + \lambda y^4, \quad \lambda > 0. \quad (10)$$

The discretized Euclidean action takes the form:

$$S_E[y] = \sum_{j=0}^{N_t-1} \left[\frac{(y_{j+1} - y_j)^2}{2\eta} + \eta \left(\frac{1}{2}y_j^2 + \lambda y_j^4 \right) \right], \quad \eta = \frac{\beta}{N_t}, \quad (11)$$

with periodic boundary conditions. The non-Gaussian nature of the local measure precludes direct selection by a Heat-Bath algorithm; updates were therefore performed using local Metropolis sampling interleaved with microcanonical overrelaxation steps [2, 3].

Data collection was completed at fixed thermal extent $\beta = 5$ for couplings in the range $0 < \lambda \leq 1$. Each observable O was extrapolated to the continuum through a fit $O(\eta) = a + b\eta^2$, consistent with the systematic error of order $O(\eta^2)$ induced by lattice discretization [1, 3]. The continuum values obtained from Path Integral Monte Carlo were validated by comparison with pseudo-analytical estimates derived from an exact diagonalization (ED) on a truncated basis, whose details are discussed in App. B.

The continuum behavior of $\langle y^2 \rangle_{\text{cont}}$ and $\langle y^4 \rangle_{\text{cont}}$ as functions of the coupling parameter λ is illustrated in Fig. 12. Both observables decrease monotonically as λ increases, showing perfect statistical agreement with the ED references. A representative example of the lattice extrapolations for the quantity $\langle y^4 \rangle$ is reported in Fig. 13.

Energy balance and virial theorem. The intercepts extracted from the scaling analysis make it possible to reconstruct the continuum mean potential value, $V_{\text{cont}} = \frac{1}{2}\langle y^2 \rangle_{\text{cont}} + \lambda \langle y^4 \rangle_{\text{cont}}$. The kinetic contribution is evaluated by invoking the quantum virial theorem $2\langle K \rangle = \langle y dV/dy \rangle$ [4], which, when applied to the potential under study, provides an expression depending only on the observed spatial moments:

$$K_{\text{vir,cont}} = \frac{1}{2}\langle y^2 \rangle_{\text{cont}} + 2\lambda \langle y^4 \rangle_{\text{cont}}. \quad (12)$$

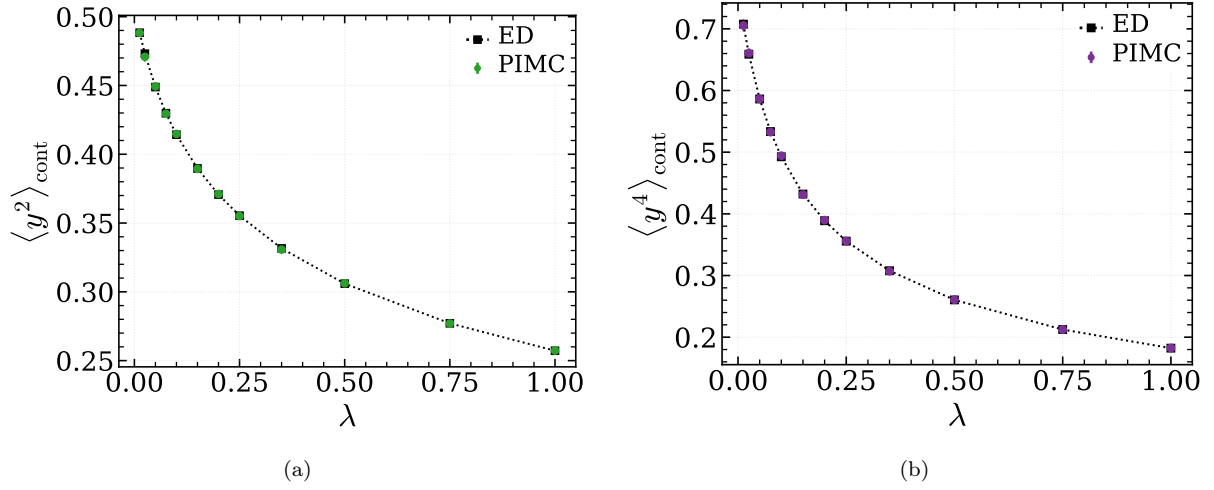


Figure 12: Continuum-limit values of $\langle y^2 \rangle_{\text{cont}}$ (a) and $\langle y^4 \rangle_{\text{cont}}$ (b) at $\beta = 5$ as functions of λ . The dotted line indicates the ED reference. PIMC errors in $[2.30 \times 10^{-4}, 3.59 \times 10^{-3}]$ (a) and $[3.46 \times 10^{-4}, 9.79 \times 10^{-3}]$ (b).

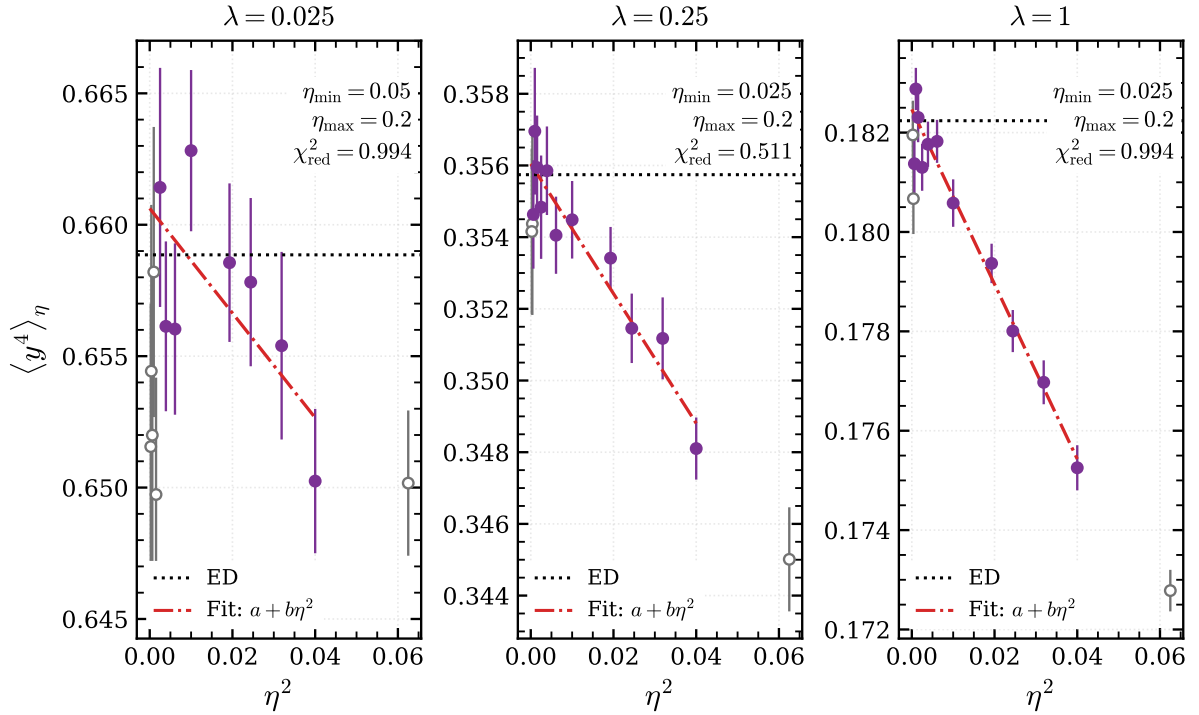


Figure 13: Asymptotic continuum extrapolation of $\langle y^4 \rangle$ for three representative values of the coupling parameter λ .

While in the harmonic limit ($\lambda \rightarrow 0$) the symmetry of the potential enforces the exact equipartition $K = V$, the presence of the quartic coupling breaks this balance by a strictly positive term equal to $\lambda \langle y^4 \rangle_{\text{cont}}$, shifting the partition in favor of the kinetic energy. This systematic deviation is confirmed by the direct comparison shown in Fig. 14, which displays the progressive and clear separation of the two energy components as the anharmonicity of the system increases.

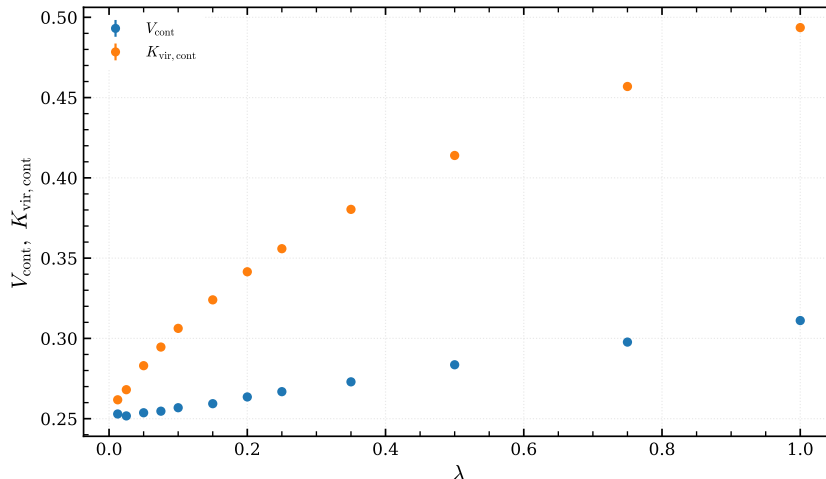


Figure 14: Continuum-extrapolated potential contribution V_{cont} and virial kinetic contribution $K_{\text{vir,cont}}$ as functions of λ .

Spatial distribution and energy spectrum. The geometric impact of anharmonicity on the wave function was assessed by reconstructing the spatial probability density $P(y)$ at $\beta = 5$. As illustrated in Fig. 15a, as λ increases the distribution becomes progressively narrower, with suppression of the asymptotic tails and a corresponding increase of the peak around the origin.

The spectral analysis was carried out by solving a generalized eigenvalue problem (GEVP): given a basis of operators $\{O_i\}$, the connected correlation matrices were constructed and diagonalized,

$$C_{ij}^{(c)}(\tau) = \langle O_i(\tau) O_j(0) \rangle - \langle O_i \rangle \langle O_j \rangle, \quad (13)$$

and the even-parity sectors $O_{\text{even}} = \{y^2, y^4\}$ and odd-parity sectors $O_{\text{odd}} = \{y, y^3\}$ were studied. The extracted energy gaps Δ_1 and Δ_2 , reported in Fig. 15(b), show a monotonic increase with λ . The ratio $R = \Delta_2/(2\Delta_1)$, shown in Fig. 15c, directly reveals the breaking of harmonic equidistance: equal to 1 at $\lambda = 0$, it exceeds unity as λ increases. The comparison with ED validates the PIMC estimates.

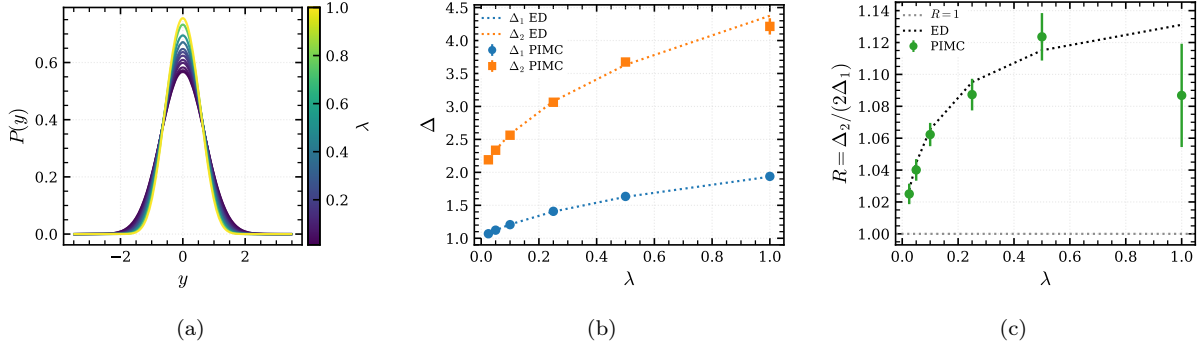


Figure 15: Spatial and spectral properties of the anharmonic system. (a) Spatial probability density $P(y)$ as the coupling λ varies. (b) Excited energy gaps Δ_1, Δ_2 . (c) Anharmonicity ratio $R = \Delta_2/(2\Delta_1)$ (right) as a function of λ , compared with the ED reference curves.

A Monte Carlo Diagnostics

To ensure the statistical reliability of the MCMC simulations, the fluctuations of $\langle y \rangle$, $\langle y^2 \rangle$, and $\langle (\Delta y)^2 \rangle$ were monitored. The normalized autocorrelation function for a generic observable O , defined as

$$C_O(t) = \frac{\langle O(s)O(s+t) \rangle - \langle O \rangle^2}{\langle O^2 \rangle - \langle O \rangle^2}, \quad C_O(0) = 1, \quad (14)$$

made it possible to quantify the temporal correlation between successive measurements [2]. The integrated autocorrelation time was estimated by truncation at a self-consistent window W according to the Madras–Sokal criterion [5]:

$$\tau_{\text{int}}(W) = \frac{1}{2} + \sum_{t=1}^W C_O(t). \quad (15)$$

As shown in Fig. 16a for a lattice with $N_t = 400$ at $\beta = 5$, the introduction of deterministic microcanonical overrelaxation steps drastically reduces the autocorrelation scales of the slow observable $\langle y \rangle$ [2]. The spectral analysis of the fluctuations, illustrated in Fig. 16b and summarized in Tab. 4, confirms that the linear coordinate is the slowest-relaxing quantity in the system.

Observable	τ_{int}
$\langle y \rangle$	122.4
$\langle y^2 \rangle$	50.2
$\langle (\Delta y)^2 \rangle$	5.25
H_{ren}	5.34

Tabella 4: Integrated autocorrelation times estimated at $\beta = 5$ and $N_t = 128$.

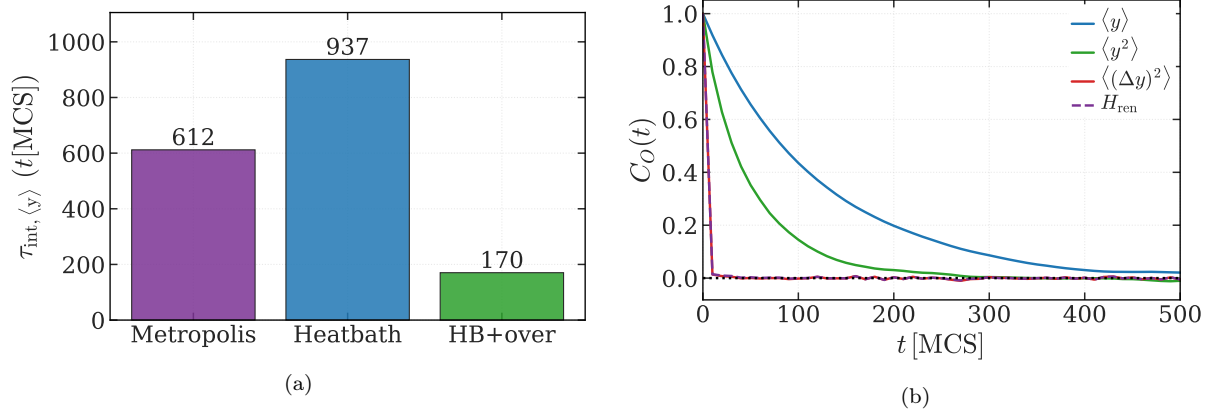


Figure 16: Monte Carlo diagnostics for the harmonic oscillator: (a) integrated autocorrelation time τ_{int} of $\langle y \rangle$, in sweep units, for different update algorithms at $\beta = 5$ and $N_t = 400$. (b) Normalized autocorrelation functions of the main observables at $\beta = 5$ and $N_t = 128$.

Thermalization of the system was validated by simulating two Markov chains with independent initializations [3]. As shown in Fig. 17, the application of the conservative criterion $n_{\text{therm}} = \max(2n_{\text{therm}}^{(0)}, 20\tau_{\text{int}})$ imposed the systematic discarding of the first $n_{\text{therm}} = 4 \times 10^5$ sweeps for each lattice configuration.

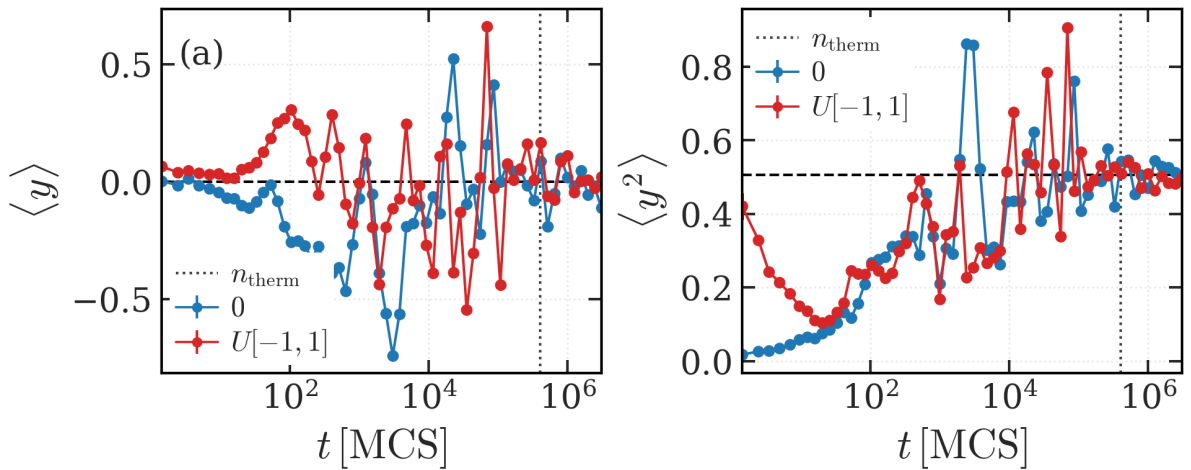


Figure 17: Thermalization diagnostics on a logarithmic scale for $\beta = 5$ and $N_t = 512$, starting from different configurations.

B Exact Diagonalization of the Quartic Oscillator

Pseudo-analytical reference results for the anharmonic system were obtained by numerical diagonalization of the dimensionless Hamiltonian $H = H_0 + \lambda y^4$ in the truncated basis of harmonic eigenstates $\{|n\rangle\}_{n=0}^{N_{\text{basis}}-1}$, defined by the condition $H_0 |n\rangle = (n + 1/2) |n\rangle$. The matrix representation takes the form:

$$H_{mn} = \left(n + \frac{1}{2}\right) \delta_{mn} + \lambda \langle m | y^4 | n \rangle. \quad (16)$$

The matrix elements of the quartic operator are expressed through standard combinations of the ladder relations $a |n\rangle = \sqrt{n} |n-1\rangle$ and $a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$. Conservation of

parity makes it possible to separate the problem into independent even and odd blocks. Truncating the basis at $N_{\text{basis}} = 160$ ensures convergence of the lowest eigenvalues within machine precision. Thermal averages for a generic operator O are computed using

$$\langle O \rangle_{\text{ED}} = \frac{1}{Z} \sum_k e^{-\beta E_k} \langle \psi_k | O | \psi_k \rangle, \quad Z = \sum_k e^{-\beta E_k}. \quad (17)$$

C Source Code

The complete source code for the simulation and analysis is available at the following public repository:

https://github.com/Phoeredor/Harmonic_Quartic_Quantum_Oscillator

Riferimenti bibliografici

- [1] Claudio Bonati. *Some notes for Metodi Numerici per la Fisica / Computational Physics Laboratory*. Lecture notes, Università di Pisa. Versione del 29 giugno 2026. 2026.
- [2] David Landau e Kurt Binder. *A Guide to Monte Carlo Simulations in Statistical Physics*. 5^a ed. Cambridge University Press, 2021.
- [3] Massimo D’Elia. *Appunti del Corso di "Metodi Numerici della Fisica Teorica / Parte III - Applicazioni al calcolo del path-integral in meccanica quantistica"*. Lecture notes, Università di Pisa. 2016.
- [4] M Creutz e B Freedman. «A statistical approach to quantum mechanics». In: *Annals of Physics* 132.2 (1981), pp. 427–462. ISSN: 0003-4916. DOI: [https://doi.org/10.1016/0003-4916\(81\)90074-9](https://doi.org/10.1016/0003-4916(81)90074-9). URL: <https://www.sciencedirect.com/science/article/pii/0003491681900749>.
- [5] Neal Madras e Alan D. Sokal. «The pivot algorithm: A highly efficient Monte Carlo method for the self-avoiding walk». In: *Journal of Statistical Physics* 50 (1988), pp. 109–186. DOI: [10.1007/BF01022990](https://doi.org/10.1007/BF01022990).